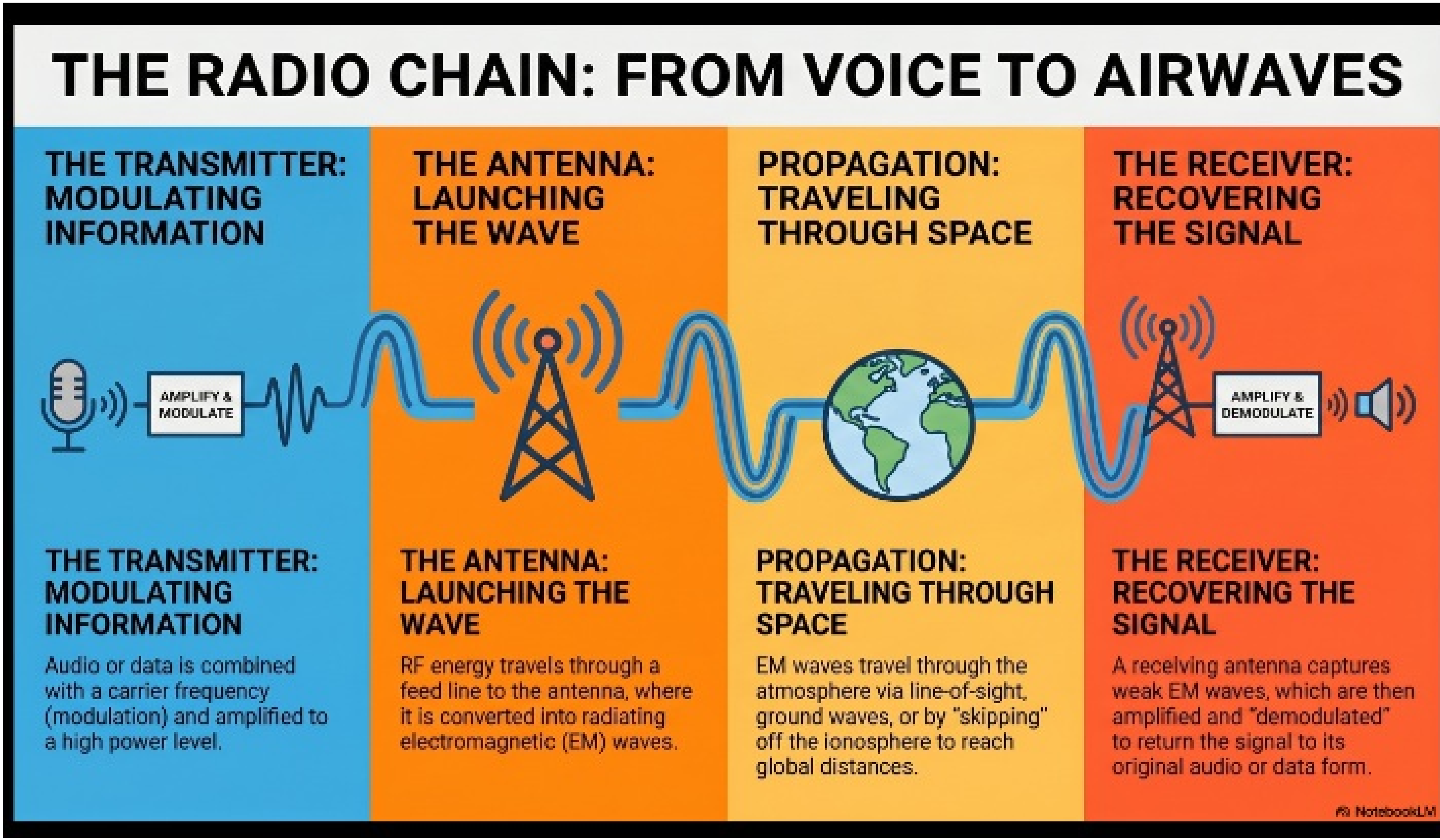
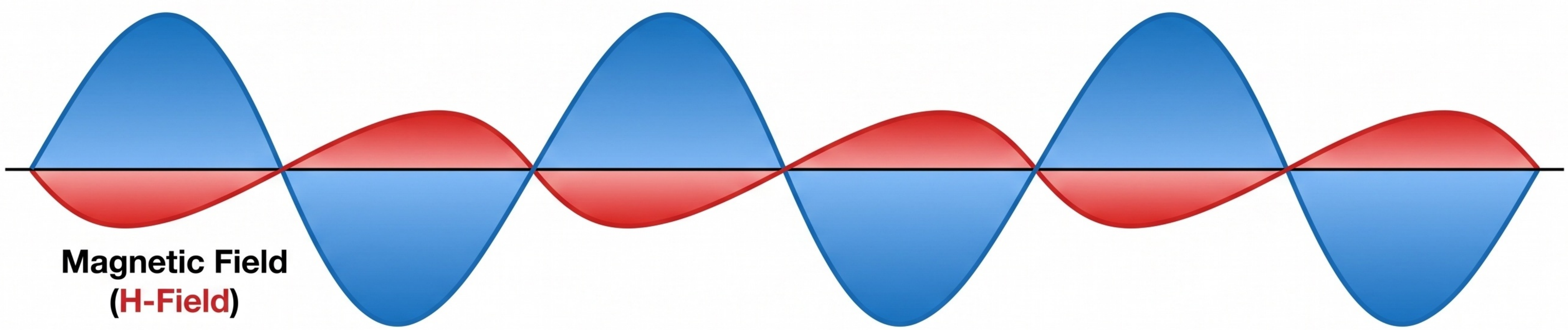
REVIEW


THE RADIO CHAIN: FROM VOICE TO AIRWAVES
THE TRANSMITTER: MODULATING INFORMATION
THE ANTENNA: LAUNCHING THE WAVE
PROPAGATION: TRAVELING THROUGH SPACE
THE RECEIVER: RECOVERING THE SIGNAL
AMPLIFY & MODULATE
AMPLIFY & DEMODULATE
THE TRANSMITTER: MODULATING INFORMATION
Audio or data is combined with a carrier frequency (modulation) and amplified to a high power level.
THE ANTENNA: LAUNCHING THE WAVE
RF energy travels through a feed line to the antenna, where it is converted into radiating electromagnetic (EM) waves.
PROPAGATION: TRAVELING THROUGH SPACE
EM waves travel through the atmosphere via line-of-sight, ground waves, or by "skipping" off the ionosphere to reach global distances.
THE RECEIVER: RECOVERING THE SIGNAL
A receiving antenna captures weak EM waves, which are then amplified and "demodulated" to return the signal to its original audio or data form.
NotebookLM

The Anatomy of an Electromagnetic Wave

Electric Field (E-Field)
An oscillating field whose orientation determines the wave's polarization (horizontal or vertical).

One Complete Cycle
The distance covered during one full oscillation is defined as the wavelength.



Velocity of Propagation
EM waves travel at approximately 300,000,000 meters per second in free space.

Transverse Propagation
Both fields are at right angles to the direction the wave travels.

One Complete Cycle
The distance covered during one full oscillation is defined as the wavelength.

The Inverse Relationship
As the frequency f, increases -> wavelength (physical length of a cycle) decreases proportionally.

Wave Math

Frequency (f)	Wavelength (λ)	Formula (λ = 300 / f)
	30.0 MHz	10 Meters
	146.0 MHz	2.05 Meters
	300.0 MHz	1 Meter

Week 2 - Electromagnetic Waves and Wavelength

40 Meter = 8. 57 MHz
7.0 - 7.3 (42 Meters)

20 Meter

Calculating Frequency and Wavelength

$$f_{MHz} = \frac{300}{\lambda} \qquad \lambda = \frac{f_{MHz}}{300}$$

What is Polarization of RF Waves?

Polarization is:
*** something to do with the orientation of the waves**

*** Horizontal Pol. - Electric Field parallel to Earth**
*** Vertical Pol. - Electric field Perpendicular to Earth**

What frequency bands are allocated to Amateurs (https://www.rac.ca/operating/bandplans/)

160M
80M
50M ???
40M
30M ***
20M
17M (WARC)
15M
12M (cclose to CB)
10M
6M
2M